

Dear Editor She and Editor Broad,

Thank you for the previous opportunity of the review process we greatly appreciate it, and we have put additional published data and used the recommended reference ZJU1 genome to redo most of our analysis. Although the esteemed Mr. Li and I greatly look forward to publishing in BMC, we acknowledge that BMC's decision will be based on thorough, procedurally sound and professional considerations. While regarding these perplexing odd comments of the third reviewer, we feel compelled to disclose some clear facts on this issue and point out his serious mistakes and the contradictory statements that are totally against the general response from other reviewers'/researchers' views, including from BMC's. We have presented it as a point-to-point file below. His one point is that one our finding has been found by the other paper. Whereas we will show some correspondence and review comments that could reflect this issue is still not well accepted by the community and back-to-back research and additional further research are reasonable.

We were pleased with the comments from the first reviewer and we had tird addressed his/her concern. Could we present it here to see if there is a chance to share them with this professional reviewer as he may feeling interesting. We also updated the scaffold-level genomes of the mallard and spot-billed to chromosome-level with miRNA and protein-coding genes annotated. Moreover, we provided a multiple-process well-structured open resource data processes toolset which is used to achieve all the results performed in this study with more advanced features and organized it into supplement section 5.8. Thus, it would provide transparent details of entire analyses. Since a lot of people have known our findings through nonpublication way, there's a possibility that some others may aim directly to reach our conclusion with made-up proving processes. While such large-scale sequencing researches often lack strict rigor, unlike mathematical proving processes, which can be a black box that can be led toward many conclusions by the author's subjective intent rather than follow the natural facts, especially when they lack experimental results, as we have. Here we will try to break such a black box.

We've include more published date and replicate most our previous analyses on ZJU1 reference genomes and the results are remain held and even more clear. We could also update most figures if possible have a new chance and any positive response would greatly promote and speed up our work on improving the paper and we are willing to use language service or try to invite a native English researcher's help to improve this paper.

Dear Dr. Liu,

Thank you very much for submitting your manuscript entitled 'Genomic analyses reveal the origin of domestic ducks and identify different genetic underpinnings of wild ducks' to Genome Biology and please accept my apologies for the long delay in getting back to you about it. It has now been seen by three referees and their comments are available below. Please note that Referee 2 didn't provide comments to authors, but raised similar concerns about the writing as Referee 3.

As you will see from the reports, Referee 1 has raised serious concerns that outdated data and

analytical methods are used for the analysis, some conclusions are supported with limited evidence. Referee 1 has requested that you need to verify the accuracy of the data and the analytical methods, and use BUSCO to assess the completeness of genome assembly and annotation. Referee 3 has raised concerns about the limited novelty and writing. In the light of these comments, I am sorry to say that we cannot offer to publish the manuscript.

Once again, we regret that we cannot be more positive on this occasion, but we hope that this feedback will be useful in deciding how to move forward with your paper.

Best wishes,

Wenjing She

Genome Biology

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Reviewer reports:

Reviewer #3: Dear Editor,

This is a work that reconstructs the domestication history of domestic ducks, in relation to mallard, spot-billed and wild ducks. Frankly, I really DO NOT enjoy reading through the work for multiple reasons: first, this is a topic that has been published for multiple times elsewhere, specifically for the domestication history (Zhang et al. 2018 GigaScience, Guo et al. 2021 Zool. Res., Feng et al. 2021 Nat. Comm., Fan et al. 2021 Genet. Selec. Evol. , Zhu et al. 2024 Poultry Sci.), or specific traits that undergo selection (Gu et al. 2023 Royal Society Open Sci., Tao et al. 2021 BMC Ecol. Evol., Yun-xiao et al. 2023 Animal. Genetics). And this is just an incomplete list. I acknowledge that this work may have larger sample sizes or better quality of data per individual, but I find it difficult to pinpoint how this work has any advanced or new conclusions compared to all the previous publications. Second, the whole paper is poorly written, for which I will detail below. In the introduction, although the authors introduced very briefly the domestication history and the remaining questions of domesticated ducks, they failed to articulate what motivated they performed this work, and how is this work compared to previous work, or if there are any alternative hypotheses. Another general problem is very often the authors described what a pipeline or a program has produced, without a clear understanding the biological meaning of the results. And there are multiple places that indicate the authors do not have a clear understanding the concept of evolution or selection.

Response: If the reviewer really doesn't want to review this paper, perhaps he/she can simply decline to review it, rather than spend four months rushing through this paper without clearly understanding the content and then rejecting it. The reviewer claims our paper didn't have any advance or new conclusions compared to all the previous publications. Whereas none of the papers the reviewer listed present similar results to ours: our paper illustrated that 1) among many wild duck subspecies, domestic ducks are more closely related to mallard and spot-billed ducks. 2) one piece of evidence for this inference is that we found the divergence time between mallard and spot-billed ducks is comparable to, or even shorter than, that between them and domestic ducks. So, the recently extensively studied genomic features of populations/subspecies during the speciation process,

genome islands, should be expected to appear and we detected it and found its extraordinary features. Those observations could provide information for understanding the driving force of such genomic features and the speciation process. Such as all df divergent peaks located in Z chromosomes, while the species-specific male plumage traits and male plumage recognition for mating are controlled by Z chromosome. And what is the state of selfish elements in this very early stage of the speciation process, which is the dominant factor and which is the consequence? 3) we not only identified recent selective sweeps in domestic lineages, more importantly, we were the first identified positive selection occurred in ancient wild species after their split from domestic lineages. This strongly implies how flightiness has evolved and whether such ability makes wild lineage survive from being domesticated. Except for one paper mentioned by the reviewer, 'Guo et al. 2021 Zool. Res', which provides similar results to one of ours in that we both put the divergence time between domestic and wild ducks to a time point before human civilization.

But even for this point, I think the third reviewer took this conclusion for granted is in sharp contrast with the previous reviewer's response. As quite a new finding, it needs more research and evidence to make all things clear. we list some previous reviewers' responses outside BMC's in the end and pick some quotes here "The text and presentation are difficult to follow and the conclusions are questionable and seem unlikely.", "Altogether it would be impossible to replicate the study with the information provided, and I cannot judge in any way whether the grounds on which the results build are solid. Given the maybe rather surprising result of domestic duck's divergence long before we would expect domestication to have taken place, I find this worrying (admittedly, it is in principle possible that domestic ducks are remnants of an else extinct lineage; but I'd have to be convinced of this).", "most serious are the strange results obtained for the estimated time of divergence between the domestic duck and the mallard and spot-billed duck. In the study by Zhou et al, the estimated time of divergence is around 2500Ya which is in line with previous estimates. The divergence time reported in the current study 100-200 Kya in combination with the estimated Ne (Fig. 2d) is Highly unlikely". This is far away from the third reviewer's claim that there seems no need for more research to supporting. And his criticisms of my professionalism on evolution are also unrealistic. Regardless of the fact that we had first published this finding in 2016 and in my Ph.D thesis titled "*Genomic Analyses Reveal the Evolution and Domestication Process of Ducks*" in China Agriculture University and 'Liu R, et al: *Genomic analyses reveal the origin of domestic ducks and identify different genetic underpinnings of wild ducks. bioRxiv 2020:2020.2002.2003.933069*.'. I once made a lot of presentations to so many professional scholars and learned from their judgments and appraising. None of them ever denied my group and me the qualifications. And without having it published, I had to apply for positions with the draft and I got feedback from renowned scientists who also affirmed that I'm qualified to do such research in evolution and genetics(Confidential info.2). Furthermore, how could we have made the finding even before the paper the reviewer mentioned, without a clear understanding of those evolutionary or genetic methods and theories at all? (b.t.w. Although we admit we need some more help to make the text better, this is the first time I can't be grateful for the reviewer's instructions and the comments are overly subjective and emotional with so many grammar errors and lack of professionalism.)

Though this not-well-accepted finding was one important reason for our submission being rejected before, I think, on the contrary, this is exactly the reason why it should be published. As we were prudent about this finding and really wanted to figure it out and once ventured to communicate this

finding and idea privately with some researchers in this area. Fortunately, they were also interested in such findings but also warning this is a provocative idea. So our expression may look cryptic. The third reviewer's comments are against all previous reviewers including those from BMC Biology recently and the first reviewer GBIO here. We are sorry we didn't include enough citations in our introduction. Because the first several submitted editions' introductions are almost about avian influenza and through the upheavals and turbulence, now rewriting I'm a little bit lost the ideas to totally express the original thoughts and all related considerations. In addition, our paper is the actually first duck whole genome sequencing research paper and now I have cited the papers that the reviewer listed.

Major comments:

1. Line 68, it is strange that the authors described here the limitation of PSMC, then in the results they performed PSMC analyses themselves. More importantly, the authors need to describe what has been reported for the domesticated ducks vs. mallard and spot-billed in the previous results using PSMC. It is also important to introduce the previously inferred divergence time between domesticated vs. wild ducks so that the readers can have a comparison.

R: As we described in our paper, we performed demographic simulations with two popular and well-used methods that are based on different rationales and can complement each other. As the reviewer said PSMC is a conventional method that has been applied in many researches. So we also performed this powerful analysis. Although this is a magnificent and sophisticated method that can utilize only one individual's genome to infer the whole species/population demographic history based on the variation in density of genetic variations across genome, and it has been validated in many research, when we use it, we must realize its limitations that it might be confounded by complex population structure and is more suitable for inferring ancient history rather than recent history. So we also performed dadi to complement and we found that, for the effective time range estimated by both simulation methods, our results are consistent.

As to the reviewer's idea to focus on the exactly divergence time, first, we have explained our logic in article that the specific divergence time is less informative as it might be influenced by many factors such as mutation rate, how many years a generation and for specifically in dadi simulation, the dilution times for sample mutations and the segment length the mutation from, so rather than focus on the specific divergence time, we find the divergence time between wild and domestic are comparable with or even longer than the splitting between the two wild populations. This is more informative than singly focus on the exact divergence time estimated by PSMC a single software. I appreciate the feedback provided, though some of these comments seem not come from a professional reviewer, rather, it gives the impression of being written by a student on behalf of a reviewer, just like what I used to do during my student periods. It's using a static and overly simplistic mindset, combined with incorrect knowledge to approaching the problems. So we dare to point the flaws of these comments. Again we have explained in paper, line 191, that PSMC can not be formally used to infer the divergence time (though MSMC can do that whereas it has been point out that less accurate than PSMC in many respect.), it can reflect the trends and the shape of PSMC curve in our paper and others are almost the same. Previous paper mostly focus on the different point from my paper, and actually we done our jobs earlier than 2016 and made a lot of public presentation and open access publish before those paper. As to 'Guo et al. 2021 Zool. Res' the

reviewer mentioned, it come to some similar points that my paper reached. But they used previous paper's data to reached completely opposite conclusions without any explanations. For example zhang et al. 2018 Giga science, which is very good works, they found 'demographic parameters estimated from the single domestication model indicated that domestication occurred 2228 years ago, with 95% CI±441years ago, followed by a rapid subsequent divergence of meat breed from the egg/dual-purpose breeds roughly 100 years after the initial domestication event.'

2. Line 102, there need to be some introductions here how these samples of domesticated ducks are collected, how representative are they?

R: Thanks for the comment. We have detailed descriptions in the Methods section and discussed their representative in supplemental information, so we didn't present it hear in the results section. We have Fig. S11 show their distribution. As we mentioned in supplemental information, domestic ducks were collected from different breed sources each time. with support by Aoji Duck Farm, Zhejiang Province and Beijing Golden Star Duck Co., LTD. Of course, we can just move all those information into the results section if we are asked to do so.

3. Line 125, there should be some introductions here regarding how the three lineages, mallard, spotted-bill and domesticated ducks are related, according to previous work. And this paragraph should end with how the new results, if any, in this work, are consistent or contradicting the published results.

R: Thanks for the reviewer's suggestion and I do believe the reviewer is very professional in presenting summaries and elegant text, especially introductions. Whereas I acknowledge that as a math and physics background researcher, I may not be good at writing a rich summary introduction on the biological context, instead we present the necessary background where our research starts. Actually, we finished this paper before all those listed papers, with public records and a lot of groups in the community know that. Here we'd like to make a formal publication and we apologize for lacking enough description on some previous duck research and thanks for the reviewer listing so many pieces of literature above. We do believe previous works did good research work and we did cite some in our paper and this edition we added some more. Whereas comparing and summarizing all those papers on ducks would be a review, we found we were focusing on different points with different methodology. Usually, we only present the previous research that we referred to in our research or discovery process and describe related results and logic. We were sorry for the lack of enough citing previously published articles. Whereas we are not writing a review and totally solving the domestic issues need a lot more research.

4. Line 150, I don't understand here, the previous sentence suggested that mallard and spot-billed divided before the wild ducks vs. domestic ducks, isnt's supposed to be that the former is deeper?

R: The mallard and spot-billed divided before the wild ducks vs. domestic ducks is the traditional pre-assumption when we start this research and it's before our finding. Our important finding are mainly from the three lineages may experience rapid radiation and the mallard and spot-billed seems to divergent even recently. This makes us tried to put the origin of the domestic lineages before the human civilization.

5. Line 154, you need to describe which result, instead of just saying what analyses (Fst tree) indicates a good representation

R: I apologize that I did not understand the reviewer's meaning. Didn't you say don't just describe what methods produced what results? I suggested that the FST-based tree result implies that our samples might be representative.

6. Line 155, "This evolutionary model could also explain the positive f3" what does this mean?

R: We explained it in the next sentence. Line 401.

7. Line 157, do you mean ancestral polymorphisms? Line 158, what does 'drafted' mean?

R: do you mean ancestral polymorphisms? No. SNP has two alleles, one ancestral and one derived. Either one allele may inherit into the descendant lineages as fixed one. what does 'drafted' mean. We apologize for the spelling of the 'drifted' as drafted, though, I think all geneticists would probably know that is the word 'drift'. Line 401.

8. Line 165, what does 'the ancestral state was polarized' mean? Do you suggest there are multiple states in the ancestral status?

R: The singular form implies just one thing, i.e., one state. We have declared it in methods that both outgroup populations are fixed as the same allele, alt or ref allele, then we can determine the ancestral information. Not multiple states.

9. Line 171, this is not clear, what are exactly the two divergence time, between wild vs. domestics or between the two wild?

R: Thanks for the reviewer's reminder, we have rephrased it. Line 171.

10. Line 178, what does this T mean? This is not the first time where the paper just presents some measurements from the pipeline or the programme, without clarifying what is the biological meaning.

R: Thanks for the reviewer's remainder, we have added a note after. Line 175.

11. Line 186, I think this estimate has been inferred by multiple published work (see my comments at the beginning) , and there need to be a comparison here.

R: Could the reviewer specify which paper and compare what? As we mentioned above, we were focused on different issues and with different logic to illustrate it. and I didn't see those papers make any comparison with each other. furthermore, we had cited them in the paper already, e.g. line 192. As I explained above, this paper ever dominated by a different author and thus now changing to the other way to write will inevitably be influenced by the inertia of and inherent previous focal points. And I have been long time focused on using genomics/genetics to discovery primates, thus lacking interaction with groups in duck, but we will be delighted to make changes as reviewers' suggestions.

12. Line 203, how is this entire paragraph related to the domestication history of domesticated ducks?

R: Our article uses sophisticated methods and theory to discover the origin of duck domestication and we found an unexpected evolutionary history of ducks that would cause a series of consequences. At first, we thought our main tasks were the place and exact time domestication of ducks happened

and whether it's from mallard or the descendent of mallard and spot-billed wild ducks. As our finding is inconsistent with traditional Our title is more like a historical consequence, a little bit not fully reflect the whole content of the paper.

13. Line 223, there does not need to be "" for the Z chromosome here.

R: Thanks for the instruction, we have modified it in paper.

14. Line 251, do you mean 'positive' selection?

R: Yes, ancient positive selection.

15. Line 254, the first several sentences do not make sense. How can one infer selection from changes of population sizes, which are two independent processes.

R: selection occurs because it grants individuals advantage traits that adjust to the environment better then if a population/subspecies becomes better than others or their past, they will more easily survive and produce more individuals i.e. population increased.

16. Line 255, 'new traits advantageous'. Also artificial selection would lead to reduction of population size, instead of 'environmental changes' described in the paper.

17. Line 262, does '150k' mean the lengths of overlapped genomic regions?

R: Yes

18. Line 262-270, the whole part between 262-270 is rather descriptive, and has nothing to do with actual biological meaning

R: we described the outline shape of those regions that showed exception signal using this method. And we acknowledge we don't have profound biological knowledge.

Reviewer's Responses to Questions

Reviewer #1: The manuscript of Liu et al. aims to present mallard and spot-billed genomes and put the origin of domestic duck lineages back to a time far beyond the human civilization, this is a very interesting and valuable research. However, some conclusions are based on relatively outdated data and analytical methods, leading to partially insufficient evidence. Given the current availability of a large amount of high-quality publicly accessible data, it is recommended to integrate these data to establish a more rigorous chain of evidence. Additionally, I have some suggestions for improving the manuscript.

1. Line 9: BGI-Shenzhen. Same problem in additional file.

R: Thanks for your concern. We apologize for this typo and we have modified it.

2. Line 29: BGI_duck_1.0 Pekin is no longer the most widely used reference genome. It is recommended to replace it with a currently higher-quality reference genome which may have better implications for reducing potential errors and biases.

R: Thanks for your suggestion. We have almost redo all the analysis as described in main text (line 102, 530, 575) and supplementary materials.

3. Line 71: Use italics for "*Anas zonorhyncha*"

R: Thanks for your instruction. We have modified it accordingly.

4. Line 99: Compared to the currently available genomes, the description of "high-quality drafts of the mallard and spot-billed genomes" needs to be toned down.

R: The reviewer is correct and we apologize for the previous overlook to modify this claim. We have removed the word 'high-quality' in line 106 and we'd also delight to mention that we have updated these assemblies to chromosome level by referring to the available wild mallard genome, with annotated protein-coding gene and noncoding miRNA of our own datasets.

5. Line 102: It should be clearly stated here how many libraries were constructed using 114 domestic duck samples, and it is important to explicitly explain the pooling sequencing strategy to avoid any potential misinterpretation. Meanwhile, there are statistical errors in Table S11. For example, SRS1419172 and SRS1419168 have different data size but same sequence depth. Data should be described carefully.

R: Thanks for the reviewer's instruction. We have modified there line 111 and further explained in methods line 554 and Additional file 1 section 4.

6. Line 111,Line 135: Propose incorporating statistical tests to enhance the credibility of the conclusions in this section.

R: I appreciate the reviewer's proposal and we have added a percentage similarity there. We also are seeking statistical tests to enhance this claim and similar claims in other places and considering related kinship issue. Perhaps in this place, the data absolute values of the entire genome, not sampling value not estimates, 99% percentage identity could illustrate this point.

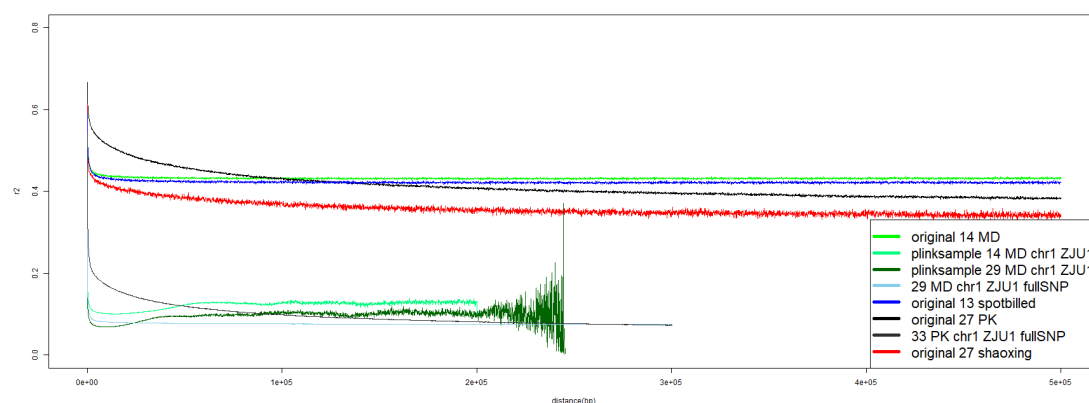
7. Line 129: Please verify the accuracy of the data and analytical methods. The LD decay results seem to be inconsistent with general recognition, indicating that the LD decay rate in domestic ducks appears to be faster than that in the mallard.

R: Thanks for pointing it out. This result does seem a little bit odd and unlike PCA or PSMC et al., has been replicated with the same results by others. But the shape of the LD decay curves seems reasonable, whereas the curves for mallard and Spot-billed are elevated overall and slightly down overall for the Pekin and Shaoxing breeds. We use the same process for new results on ZJU1 (chr1 for efficiency) and the wild ones shift downward overall. It seems due to the scaffold fragment issues: The LD decay curve for the genome with a scaffold level may be higher overall than the chromosome level, but the shape is essentially the same. We then utilized all SNP sets without dilution using PopLDdecay and found the curve shifted downward overall once again. The curve for mallard we tested will even slightly shift downward again when we incorporate 15 individual mallard duck samples which are 29 mallard in total. This is a little bit like the sample size effect in PCA that we have taken care of(5.4.1 first paragraph).

The explanation for LD decay seems complex and besides the measurement error, it is related to many factors. The rapid decay is consistent with the high recombination rate of ducks estimated from old data sets of Pekin ducks, data not shown. The artificial selection or inbreeding may cause the evaluated LD in a shorter distance, i.e. slower decay as the figure shows. Although wild

populations usually have a bigger population size that would present a lower LD overall, the population stratification would probably evaluate the LD overall. The measurement of N_e may also be influenced by factors such as population structure and mating system. Regardless of the bias, Watterson's thetas of Pekin and Shaoxing are almost half of the mallard and spot-billed ones' and dadi presents Watterson's theta values of 1228, 1062 for Shaoxing, and Peking compared to ~2500 for both wild species, is also half of the wild ones. Whereas we lack the actual population size information nor do we specifically achieve the accurate value. And we only refer to LD decay to consider the window size and dilution ratio. So, for the sake of rigor and not obscuring our main points, we have removed it from the current version since LD decays do not consist of our evidence chain. We appreciate you pointing out this interesting thing again.

We used in-house scripts to randomly select SNPs at a dilution factor of 0.1 and invoked plink with the parameter '--ld-window-r2 0 --r2 --ld-window-kb 500 --ld-window 99999'. And the script 'calculateLDUsePlinkForScaffold.py' and 'binld.py' that were used to produce this result have been uploaded to github.com/ruiuiacd/atcgmagic. Due to the work position shift, the original files have been lost, but one can absolutely and completely replicate the result with the data and scripts I provided.



Our original LD decay on the above part of this figure as shown in our previous paper are from the uniformly random sampling to 0.1 times then run plink with 'ld-window 99999' to incorporate all sampled SNPs. Whereas the plink with '--ld-window-kb 500/200 --ld-window 500/800' sample nearest SNPs first and less sufficient to cover a long distance that will make the curve wave.

In summary, the sample size, fragment size (scaffold/chromosome) and the way to sample SNPs will also influence the LD decay, besides the natural factors such as population structure, N_e , inbreeding and selection. In addition, our previous LD decay analysis included almost all scaffolds, encompassing those from both autosomes and sex chromosomes.

8. Line 524: Recommend utilizing BUSCO to assess the completeness of genome assembly and annotation.

R: Thanks for the reviewer's instruction, we have done the BUSCO assessment and uploaded it to <https://github.com/ruiuiacd/duckdomestic1>.

9. Line 549: As mentioned here, the different sequence strategy has resulted in several in many analytical methods, making it challenging for others to reproduce. The article should clarify the reasons for the need for special treatments, such as why reads are randomly sampled at 3X and

12X, how to handle pooled sequencing data, and why it is necessary to control the equal number of samples for each breed in PCA analysis, among other considerations.

R: Thanks for pointing it out. First, we have redone most analyses with new individual sequenced public data. And the results remain held and even better. If we have further opportunities, we can update those figures and results. Second, this time if I may submit, we have uploaded all necessary code, commands and scripts. See confidential info in this file and additional file 1 section 5.8. Theoretically, We can immediately provide any requested intermediate data run this time and show how it was reached. Almost all the procedures would be transparent. Therefore, we have revised the wording here line 559. Third, We had dealt with these issues in various places in original submission, for example, Additional file 1 Section 5.4. The reason the article might seem somewhat convoluted is that we considered many detailed aspects. But it's really hard. The reason for using these data is explained in Additional file 1 Section 4. While it might have been possible to present a polished paper using fewer and more streamlined data, we aimed to faithfully present broader datasets to better reflect the complete picture of the facts. Anyway I believe that the results and findings that we reached will be discovered by someone others in different ways and some has happened like the paper 3rd reviewer mentioned that used to reject ours. And I also know some others. reviewer ever provoke current situation like. Mostly, we used the VCF files generated from variant calling for each population separately. We never performed joint variant calling across individuals or pooled data. Actually we reached the same result without those pooling data. Such as we did dadi analyses again and aging, for only individual sequenced data, our original data without public data on the ZJU1 reference, and only our original individual sequenced data respectively, and reached the same results. But if we present it in that way, the paper will be more redundant.

10. Line 553-556: These sentences are more appropriate to be placed in the results section.

R: The reviewer is right. This is about the shape of the data. The results section appears overly bloated and if we have the opportunity to resubmit, we will consider reorganizing and refining.

11. Line 580: Suggested to include version numbers and references for all software, not limited to the mentioned admixture in this line.

R: Thanks for the reminder, we have added it here line 588. We will remember to check it over again before resubmitting.

12. Line 628: The notation of ZHP and similar terms should be standardized and consistent throughout the entire document.

R: Line 639. We have modified it there and checked throughout the document. Thanks for the reviewer's careful guidance.

13. Additional_files Line 932: Dose "Black cycles" means "Black circles" here?

R: Yes, we are greatly grateful for your correction.

14. The formatting and quality of some tables and figures in the manuscript need improvement.

R: Thank you for pointing out the need for improvements in the formatting and quality of the tables and figures. As we mentioned in the letter, we are seeking to an opportunity to resubmit, and we will try to update many figures and tables and making significant enhancements to improve their

clarity, resolution, and overall presentation.

15. Table S7. The percent of repeats (7.31% & 10.71%) is significantly smaller than the general value (~17%) in ducks. The methodology may need improvement.

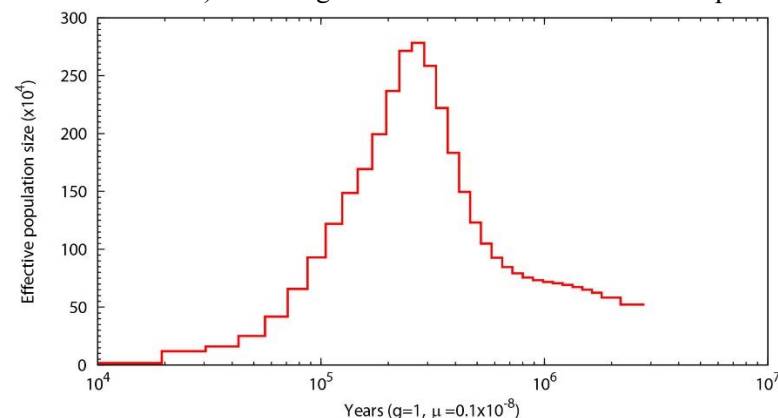
R: Thanks for the reviewer's very professional instruction. We will check for methodology and address it thoroughly in the revised manuscript.

16. Table 1. Check for possible outlier in the last column.

R: We checked our original records and found that this value is unavailable. Perhaps we subsequently use different methods to get this SNP number for the mixed one. We will modify it in the revised manuscript.

17. The accuracy of PSMC is very dependent on the reference genome. As mentioned earlier, it is highly recommended to use up-to-date, high-quality genome. PSMC analysis is also very sensitive to the sequencing depth, as displayed in Figure S19. Therefore, The ID and sequencing depth of individuals involved in the PSMC analysis (Figure 2d) need to be described in detail.

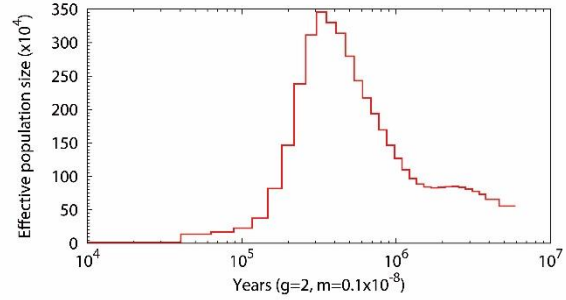
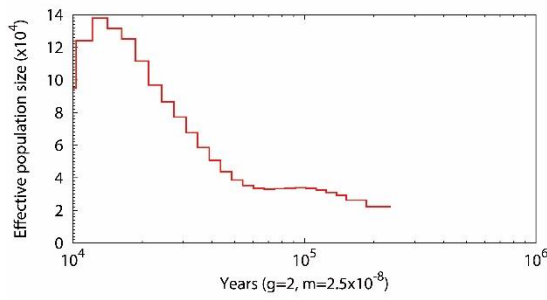
R: Thanks for the reviewer's very professional point out. We acknowledge that psmc utilizes information on the heterozygosity density across the genome and we referred some other reports performed on scaffolds (Krystyna Nadachowska-Brzyska et al. 2015). So we performed it. Now we have results on chromosomes level and the results remain unchanged (Shaoxing show exception described below). If we got chance to resubmit we will put it in. here we presented one.



Above figure is psmc curve of Pekin 44487_36 using ZJU1 genome as reference.

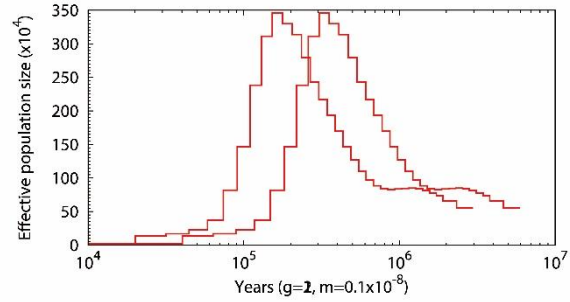
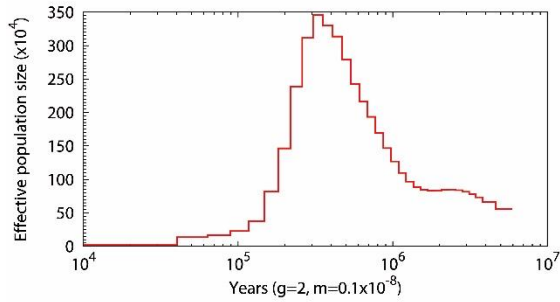
We had taken care about the depth's and other factors' influence in advance. Beside the depth's influence we presented in this paper, we also had some experiments on generation, mutation rate parameters and unique mapped vs all mapped, all scaffolds vs 40k above scaffolds' influence. We presented here (Pekin 41224_42 as an example):

Changing mutation rates

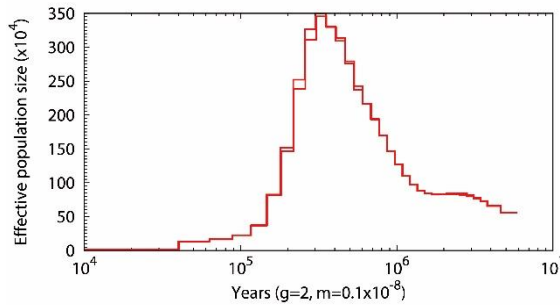


40k_plus vs all scaffolds(overlap totally)

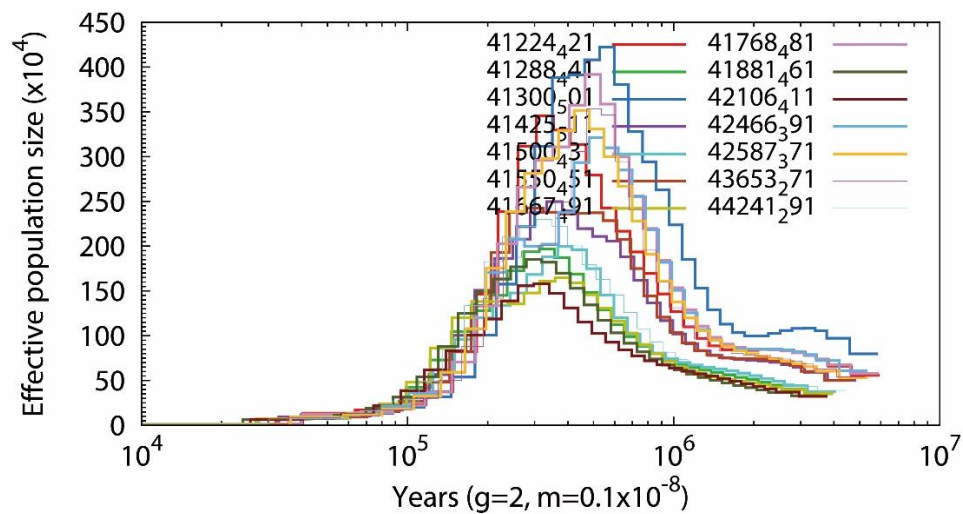
Changing years per generation $g=1$ VS $g=2$

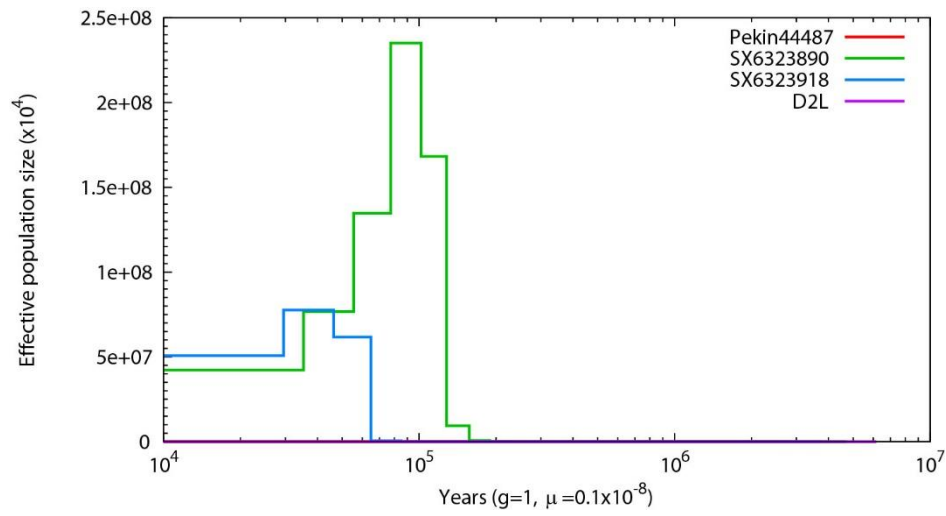


Unique mapped vs all mapped (almost the same)



As we considered the reader may concern this important issues, so the IDs were marked in the supplementary figures such as Fig. S19. Whereas, the text is slightly distorted. Here we present more





This figure shows that the Shaoxing samples SX6323890, SX6323918 with ~ 8 still with exception performance as we previously described in the paper.

18. In section 5.4 in additional file, there are a lot of misquotes of figures. Such as line 407, line 410, line 417, line 425 and so on. Please check it carefully.

R: We appreciate the reviewer picking it out and apologized for previous' overlook. We have modified them and check overall.